

End-to-End Sales & Customer Analysis

(Excel, SQL & Power BI)

1. Executive Summary

This project focuses on analyzing sales performance, profitability, customer behavior, and operational efficiency using Excel, SQL, and Power BI. The objective was to identify key business drivers, uncover profit leakage areas, and generate actionable recommendations for improving revenue and profitability.

The analysis revealed strong sales performance from the Technology category, significant regional disparities in profitability, the negative impact of excessive discounting, and several high-revenue products that consistently generated losses. Interactive dashboards were developed in Power BI to support business decision-making.

2. Business Problem

Organizations often focus on increasing revenue without understanding the factors affecting profitability. Management needed answers to the following questions:

- Which categories and products generate the highest revenue?
- Which areas contribute most to profit?
- How do discounts impact profitability?
- Which customer segments drive sales and margins?
- Which products generate losses despite strong sales?
- Which regions require operational improvement?

The goal was to convert raw transactional data into meaningful business insights.

3. Dataset Overview

Source: Kaggle

Dataset Type: Retail Sales Transaction Dataset

Dataset Size: Approximately 10,000 transaction records

Total Attributes : 21

Key Attributes:

- Order Information
- Customer Details
- Product Information
- Category and Sub-Category
- Region
- Sales
- Profit
- Discount

Dataset Statistics:

- Total Records: ~10,000
- Total Columns: 21
- Time Period: 2014–2017
- Categories: 3
- Customer Segments: 3
- Regions: 4

4. Data Preparation & Normalization (Excel)

The raw transactional dataset was initially analyzed and prepared using Microsoft Excel.

Activities Performed:

- Removed duplicate records
- Validated data consistency
- Standardized column formats
- Checked missing values
- Reviewed business dimensions

To improve data organization, the transactional dataset was normalized into separate structured tables.

Normalized Structure:

- Orders Table
- Customer Table
- Product Table

This structure reduced redundancy and prepared the datasets for SQL-based analysis.

5. SQL Analysis

SQL was used to perform exploratory and business-focused analysis.

Analysis Included:

- Total Sales Analysis
- Profitability Analysis
- Category Performance Analysis
- Customer Segment Analysis
- Regional Analysis
- Product Performance Analysis
- Discount Impact Analysis

SQL Techniques Used:

- Aggregations
- GROUP BY
- ORDER BY
- CASE Statements
- Joins
- Filtering
- Business Query Development

The SQL analysis provided the foundation for dashboard design and business insights.

Key SQL Business Questions Solved:

- Top revenue generating products
- Most profitable categories
- Region-wise sales and profit analysis
- Customer segment performance
- High sales but loss-making products
- Impact of discount on profitability
- Profit margin comparison across regions

6. Power BI Dashboard Development

An interactive Power BI dashboard consisting of eight analytical pages was developed.

Dashboard Sections:

1. Executive Summary
2. Sales Trend Analysis
3. Category Performance Analysis
4. Regional Performance Analysis
5. Customer Segment Analysis

6. Discount Impact Analysis
7. Top Product Drivers
8. Loss Analysis

The dashboard enables stakeholders to monitor business performance from multiple perspectives.

KPIs Developed:

- Total Sales
- Total Orders
- Total Profit
- Profit Margin %
- Regional Profitability
- Category Contribution %
- Segment-wise Performance
- Product Profitability Analysis

7. Key Findings

Sales Trend Analysis

- Sales and profit demonstrate a positive long-term trend with significant spikes during fourth-quarter periods, indicating strong seasonal demand.

Category Performance

- Technology is the strongest performing category in terms of both revenue and profitability.
- Furniture contributes considerable sales volume but generates significantly lower margins.

Regional Performance

- The West region contributes the highest sales and profit.
- The Central region consistently underperforms and records negative profit margins.

Customer Segment Analysis

- Consumer customers generate the highest sales volume.
- Home Office customers deliver the strongest profit margin percentage.

Discount Impact

- Profitability declines sharply when discount levels exceed approximately 40–50%.
- Aggressive discounting contributes significantly to overall losses.

Product Analysis

- Several products generate strong revenue but remain loss-making, suggesting pricing or discount management issues.

8. Business Recommendations

- Introduce approval-based discount policies beyond 40%.
- Increase focus on high-margin Technology products.
- Review pricing structures of loss-making products.
- Investigate operational inefficiencies in the Central region.
- Improve product mix toward higher-profit categories.
- Monitor discount effectiveness using profitability-based KPIs rather than sales volume alone.

9. Business Impact

- This analysis helps organizations:
- Reduce profit leakage
- Improve discount governance
- Increase profitability
- Optimize product strategy
- Strengthen regional performance
- Support data-driven decision making

10. Conclusion

The project successfully transformed raw retail sales data into actionable business insights through a complete analytic workflow involving Excel, SQL, and Power BI.

The findings highlight the importance of balancing revenue growth with profitability, controlling excessive discounting, and focusing on high-performing products, regions, and customer segments to drive sustainable business growth.